

ICT5253 Cloud Architectures and Solutions

Assessment 3: Cloud Design and Deployment

Assessment Overview	
Weighting	30% of unit total
Due Date	Week 12
Assessment Type	Group (maximum of 2 students)
Word Count / Length	3000 words +/- 10%
Unit Learning Outcomes	In this assessment, you will be tested on whether you have successfully met the following Unit Learning Outcomes (ULOs): • ULO1: Critique the fundamental principles and paradigms of cloud computing architectures. • ULO2: Appraise recent technology innovations that enable effective cloud computing. • ULO3: Evaluate the security issues and challenges in cloud computing. • ULO4: Critically review the technological and ethical issues of cloud solutions. • ULO5: Design and implement cloud solutions to enable specific business objectives.
Submission Type and Required Format	The type of assessment you will be completing is to create and configure a simple cloud service using AWS Academy according to specified business requirements. It should include the following: 1. Report 2. Lab completion activities 3. Lessons Learned Log
Assessment Details	
Assessment Purpose	The purpose of this assessment is to assess students' ability to launch, resize, manage, and monitor an Amazon EC2 instance.

Assessment Instructions	In this assessment, you will need to complete the following steps: 1. Produce a report 2. Create and configure a simple cloud service using AWS Academy according to specified business requirements 3. Create a Lessons Learned Log to assess your individual teamwork performance (see sample below)
	In this assessment, you will need to complete the following steps: 1. Produce a report 2. Create and configure a simple cloud service using AWS Academy according to specified business requirements 3. Create a Lessons Learned Log to assess your individual teamwork performance (see sample below) TASK 1 Write a report discussing the following topics: 1. Contemporary Cloud Architecture a. Cloud deployment models i. Public cloud ii. Private cloud iii. Hybrid cloud iv. Briefly describe at least 3 organisations' who successfully deployed a public, private and hybrid cloud models b. Cloud service models i. IaaS ii. PaaS iii. SaaS iv. Briefly discuss at least 3 organisations' who offer IaaS, PaaS and SaaS solutions 2. Cloud Innovations a. Virtualisation vs Containerization b. Briefly describe at least 2 organisations' who successfully deployed virtualisation and containerization using Kubernetes for their cloud computing solutions 3. Current issues, challenges, and emerging trends in cloud computing 4. AWS Cloud Security a. AWS Shared Responsibility Model i. Security "of the" cloud ii. Security "in the" cloud

- b. AWS Identity and Access Management (IAM)
 - i. IAM users, groups and roles
 - ii. Security credentials in IAM
- c. Securing data on AWS
 - i. Encryption of data *at rest*
 - ii. Encryption of data *in transit*
 - iii. Securing Amazon S3 buckets and objects
- d. AWS compliance programs
 - i. Certifications
 - 1 ISO27001/27017/27018
 - 2 ISO/IEC 9001
 - ii. Laws, regulation and privacy
 - 1. EU General Data Protection Regulation (GDPR)
 - 2. HIPAA
 - iii. Alignments and frameworks
 - 1. NIST Cybersecurity Framework
 - 2. PCI DSS
 - 3. Cloud Security Alliance (CSA)
- e. Recommendation
 - i. Best practices to secure an AWS account
- 8. Conclusion
- 9. List of References
- 10. Appendix
 - a. See Task 2 Case study/Lab practical
 - i. Screenshots
 - ii. Short-answer questions
 - b. Lessons learned log (see below)

SAMPLE LESSONS LEARNED LOG

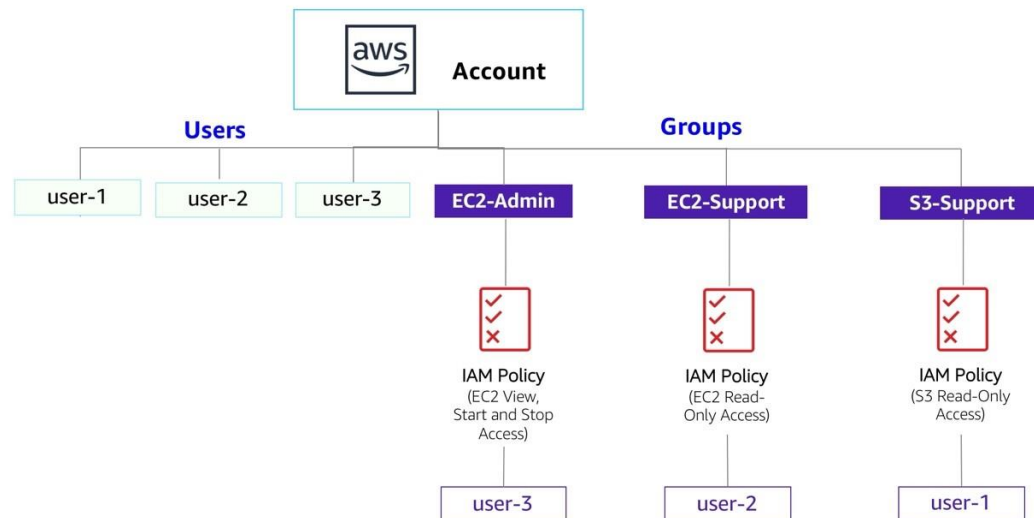
Project Name:		<i>Implement an Amazon EC2 instance</i>				
Group Members:	Student ID	Last Name	First Name	Comment	Tasks Responsible	
	<i>12345</i>	<i>Patel</i>	<i>Kushwant</i>	<i>Group Leader</i>	<i>1,2,3,4</i>	
	<i>12346</i>	<i>Shrestha</i>	<i>Sujan</i>		<i>5,6,7,8,9</i>	
No	Type	Logged By	Description	Resolution	Status	Remarks
<i>1</i>	<i>Non-technical</i>	<i>K. Patel</i>	<i>Grouping issue; group can't decide day/time to work on assignment together due to work commitments</i>	<i>Collaborate using Zoom</i>	<i>Resolved</i>	
<i>2</i>	<i>Technical</i>	<i>S. Shrestha</i>	<i>Can't log in to AWS</i>	<i>Reported issue to lecturer; account reset</i>	<i>Resolved</i>	
<i>3</i>	<i>Technical</i>	<i>S. Shrestha</i>	<i>Unknown issue on AWS preventing launch of instance</i>	<i>Reported issue to lecturer</i>	<i>Unresolved</i>	<i>Lecturer looking into it</i>
<i>4</i>	<i>Non-technical</i>	<i>K. Patel</i>	<i>Team member J. Singh uncontactable; missed 3 Zoom meetings; not answering phone; not attending classes</i>	<i>Reported issue to lecturer</i>	<i>Resolved</i>	<i>Lecturer informed group that Team member J. Singh dropped out of the class. Currently looking for replacement</i>
<i>5</i>	<i>Technical</i>	<i>S. Shrestha</i>	<i>Unknown issue on AWS preventing launch of instance</i>	<i>Reported issue to lecturer</i>	<i>Resolved</i>	<i>Wrong account used by student</i>

TASK 2 Case study/Lab practical

Your company is growing its use of Amazon Web Services and is using many Amazon EC2 instances and a great deal of Amazon S3 storage. You wish to give access to new staff depending upon their job function:

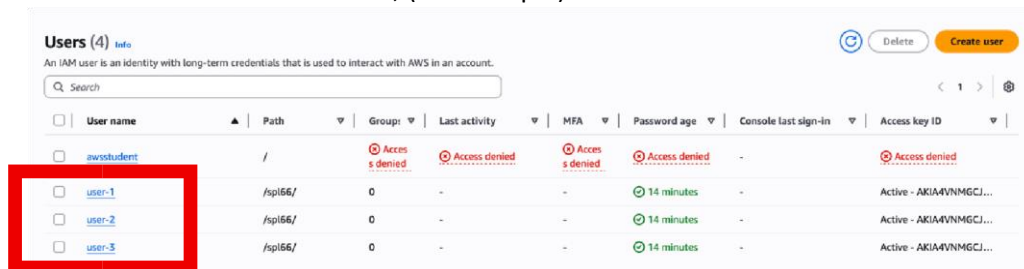
User	In Group	Permission
user-1	S3-Support	Read-Only access to Amazon S3
user-2	EC2-Support	Read-Only access to Amazon EC2
user-3	EC2-Admin	View, Start and Stop Amazon EC2 instances

This lab will demonstrate:



Required Evidence:

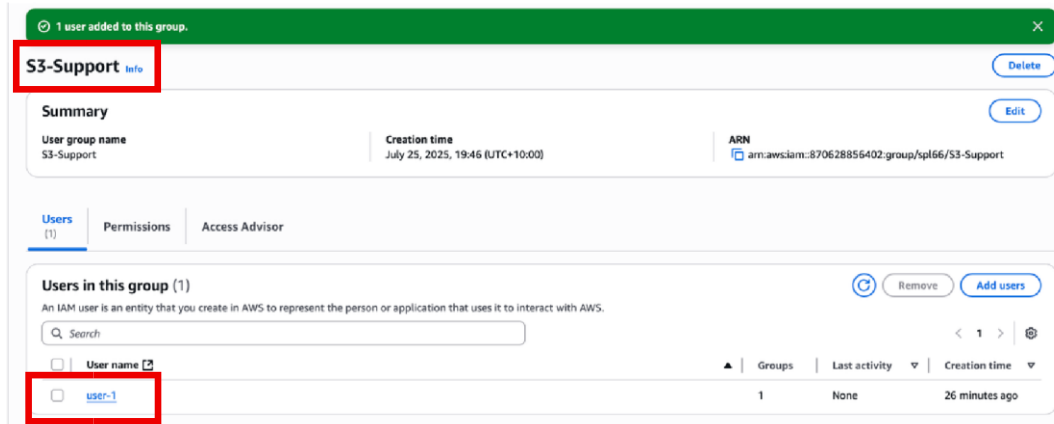
1. A screenshot of all 3 IAM users, (for example):



Answer the following questions:

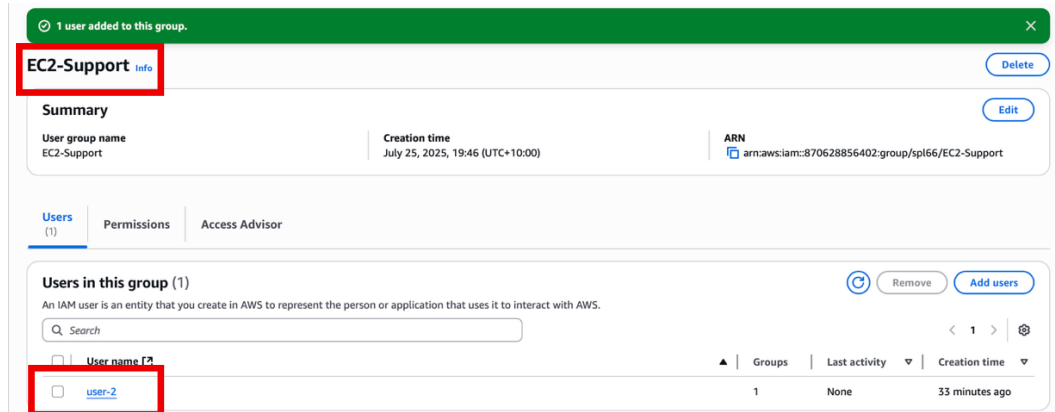
- a. What can we use AWS Identity and Access Management (IAM) for?
- b. What is the difference between an IAM User and an IAM Role?
- c. What is the benefit of enabling *identity federation*?

2. A screenshot of one of the IAM users added to the S3-Support group, (for example):



The screenshot shows the AWS IAM console for the 'S3-Support' group. A green banner at the top indicates '1 user added to this group.' The group name 'S3-Support' is highlighted with a red box. The summary section shows the user group name as 'S3-Support', creation time as 'July 25, 2025, 19:46 (UTC+10:00)', and ARN as 'arn:aws:iam::870628856402:group/spl66/S3-Support'. The 'Users' tab is selected, showing a table with one user, 'user-1', which is also highlighted with a red box. The table columns are Groups, Last activity, and Creation time.

3. A screenshot of one of the IAM users added to the EC2-Support group, (for example):



The screenshot shows the AWS IAM console for the 'EC2-Support' group. A green banner at the top indicates '1 user added to this group.' The group name 'EC2-Support' is highlighted with a red box. The summary section shows the user group name as 'EC2-Support', creation time as 'July 25, 2025, 19:46 (UTC+10:00)', and ARN as 'arn:aws:iam::870628856402:group/spl66/EC2-Support'. The 'Users' tab is selected, showing a table with one user, 'user-2', which is also highlighted with a red box. The table columns are Groups, Last activity, and Creation time.

Answer the following questions:

- What are Managed Policies?
- What happens when a policy is updated?
- What permissions are granted in an **AmazonEC2ReadOnlyAccess** policy?
- Discuss the basic structure of the statements in an IAM Policy.

4. A screenshot of one of the IAM users added to the EC2-Admin group, (for example):

1 user added to this group.

EC2-Admin [Info](#)

[Delete](#)

Summary [Edit](#)

User group name: EC2-Admin Creation time: July 25, 2025, 19:46 (UTC+10:00) ARN: arnaws:iam::870628856402:group/spi66/EC2-Admin

[Users \(1\)](#) [Permissions](#) [Access Advisor](#)

Users in this group (1) [Remove](#) [Add users](#)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name	Groups	Last activity	Creation time
☐	user-3	1	None	37 minutes ago

Answer the following questions:

- a. What is the difference between a Managed Policy and an Inline Policy?

5. A screenshot of all 3 IAM users signing in to the IAM account console, (for example):
user-1

New sign in Multi-session disabled English



IAM user sign in

Account ID or alias (Don't have?)
870628856402

☐ Remember this account

IAM username
[user-1](#)

Password

☐ Show Password [Having trouble?](#)

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user-2

New sign in ▾Multi-session disabled ▾English ▾



IAM user sign in ⓘ

Account ID or alias ([Don't have?](#))

☐ Remember this account

user-2

••••••••

☐ Show Password [Having trouble?](#)

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user-3

New sign in ▾

Multi-session disabled ▾

English ▾



IAM user sign in ⓘ

Account ID or alias (Don't have?)
870628856402

☐ Remember this account

IAM username
user-3

Password
••••••••

☐ Show Password [Having trouble?](#)

Sign in

Sign in using root user email

[Create a new AWS account](#)

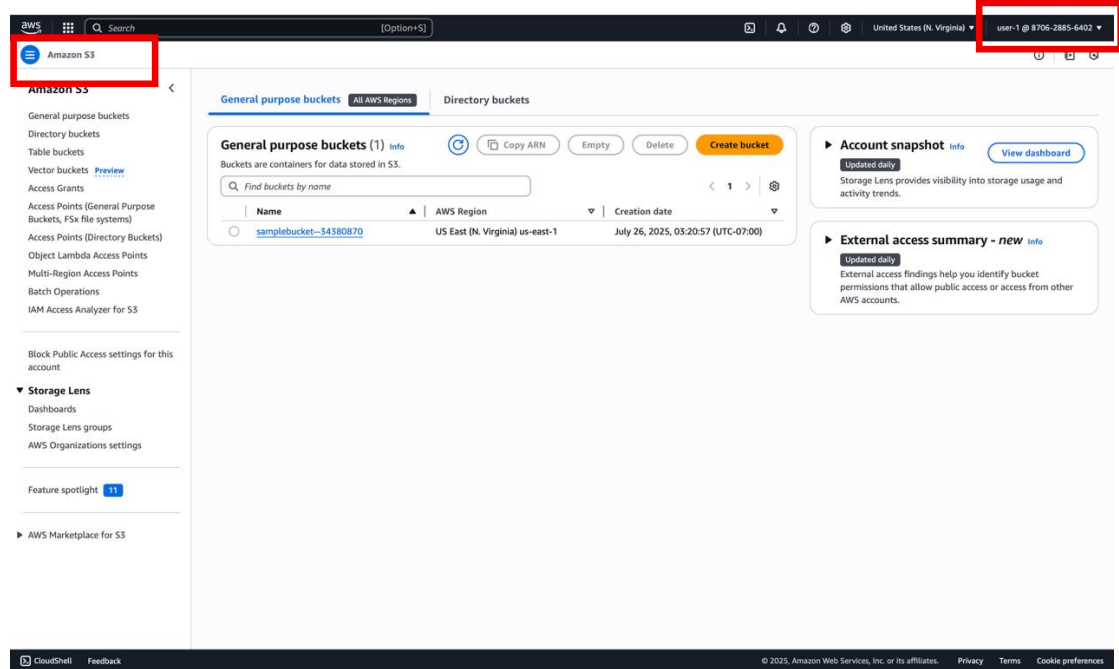
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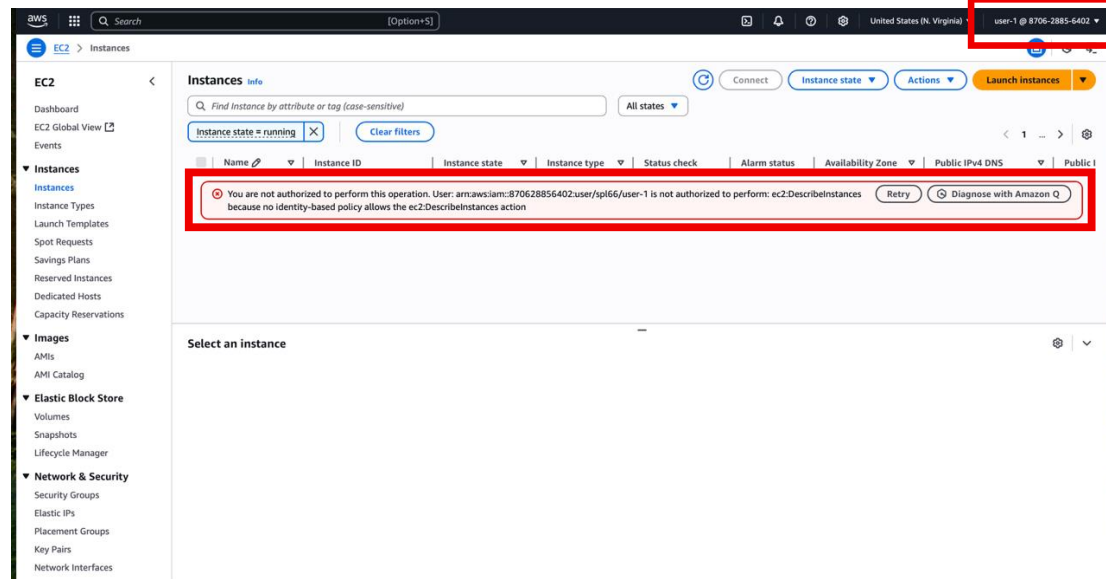
6. A screenshot of the IAM user given the S3-Support role able to access the S3 console, (for example):



Answer the following questions:

- a. Is the IAM user able to view a list of Amazon S3 buckets and their contents? Why?

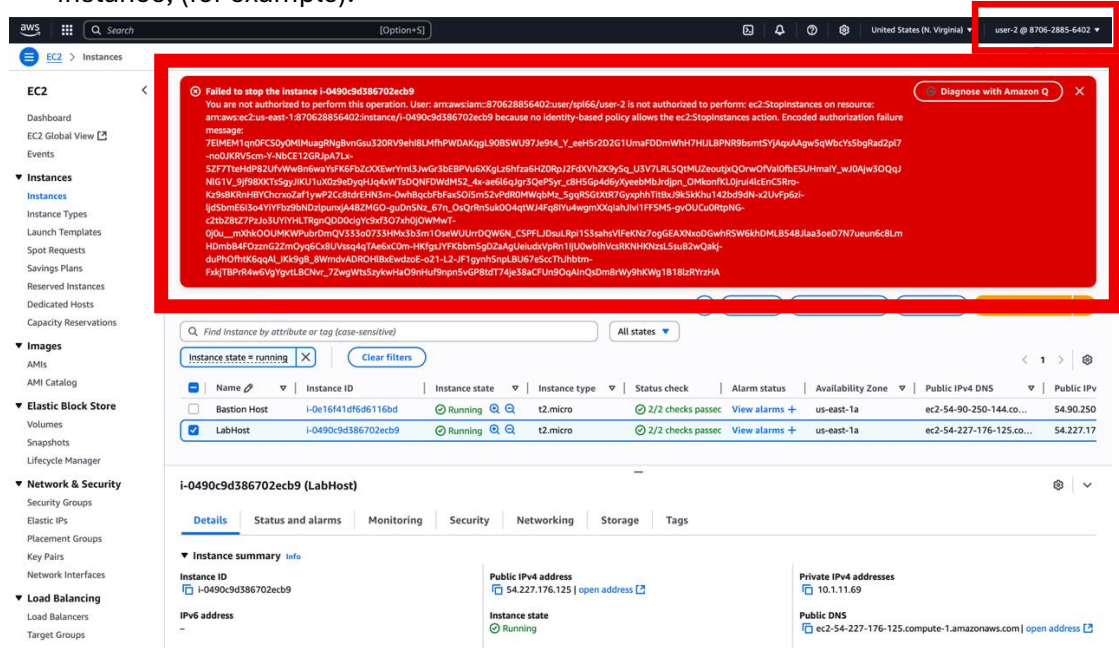
7. A screenshot of the IAM user given the S3-Support role unable to view the EC2 instances, (for example):



Answer the following questions:

a. Why can't the IAM user be unable to view EC2 instances?

8. A screenshot of the IAM user given the EC2-Support role unable to stop an EC2 instance, (for example):



The screenshot shows the AWS IAM console interface. A red box highlights an error message that appears when attempting to stop an EC2 instance. The error message states: "Failed to stop the instance i-0490c9d386702ecb9. You are not authorized to perform this operation. User: arn:aws:iam::870628856402:user/sp66/user-2 is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:870628856402:instance/i-0490c9d386702ecb9 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: ...". Below the error message, a table lists EC2 instances. The instance i-0490c9d386702ecb9 (LabHost) is shown in a 'Running' state. The console also displays details for this instance, including its ID, name, state, and various addresses.

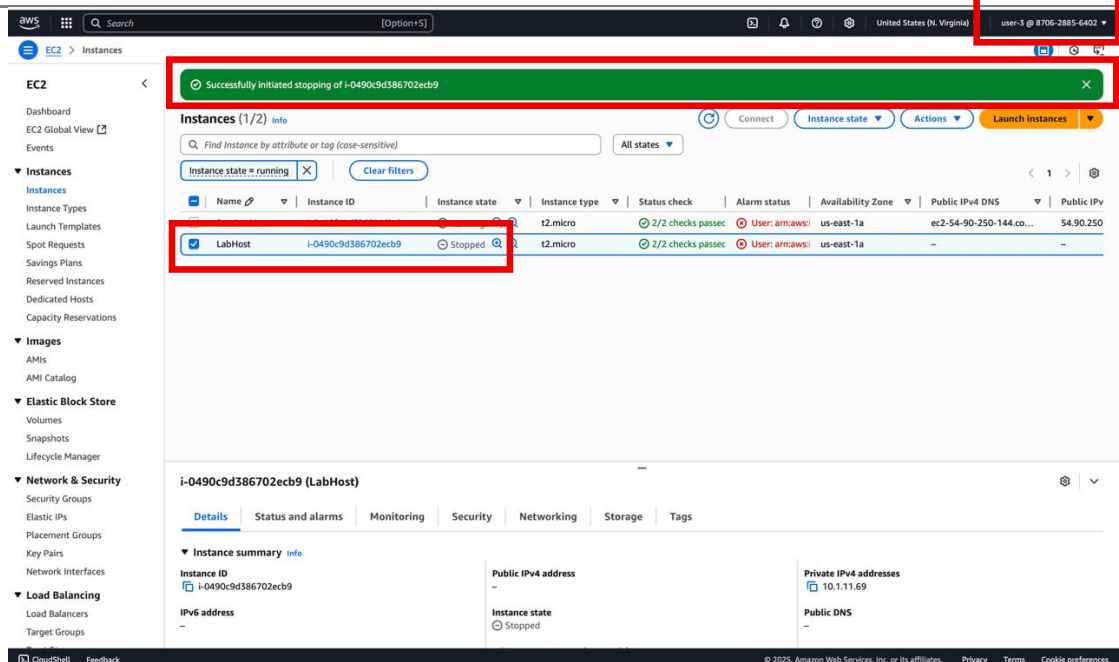
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv
Bastion Host	i-0e16f41dfe6116bd	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-54-90-250-144.co...	54.90.250
LabHost	i-0490c9d386702ecb9	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-54-227-176-125.co...	54.227.17

Answer the following questions:

a. The IAM user was able to view EC2 instance, why?

b. Can the IAM user be able to make changes to the EC2 resources as well? Why?

9. A screenshot of the IAM user given the EC2-Admin role successfully stopping an EC2 instance, (for example):



Answer the following questions:

a. The IAM user was able to stop the EC2 instance, why?

Artificial Intelligence (AI) Use	In this assessment, the use of generative artificial intelligence (AI) tools is PERMITTED. You can use AI tools to assist you in your research tasks only. You CANNOT use AI tools to generate your full assessment. Example tools might include: - ChatGPT - Copilot Students take full responsibility for the content of their assessments, and AI tools should be used as a supplement to your own research and analysis rather than as a replacement. If you use AI tools in an unethical or irresponsible manner, such as copying AI-generated output without checking it against reliable sources, you risk committing academic misconduct. Any use of AI must be appropriately acknowledged in the AI declaration on the Assessment Cover Sheet. Refer to the Using AI page on the ECA Library and Learning Support website for further guidance.
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Other Important Information

Assessment Resources	It is strongly recommended that you use the following to find academic peer-reviewed sources of information. • ECA Library catalogue and databases • Course Details for Cloud Architecture and Solutions (ICT5253 Course Reserves) • Subject Guide: ICT5253 USG For in-text citation and referencing, follow the ECA HE Student Guide to APA 7th Referencing.
Submission Requirements	On completion of your assessment: • Follow the Assessment Presentation Guidelines • Submit your document as a PDF or MS Word document via the assessment submission link in Canvas. • Include a completed Assessment Cover Sheet, including the AI Declaration, and attach it to the assessment. • Save drafts of your work.
Assessment Support	For academic support or feedback on a draft of your assessment, please email academic.support@eca.edu.au For assistance with finding resources, such as books and journal articles, please email library@eca.edu.au For information and guides on tackling assessments and developing your academic skills, please visit in the ECA Library and Learning Support website: https://eca.libguides.com/ For queries about this specific assessment task, please contact the Unit Coordinator.

Assessment 3 Rubric

Criteria (weighted as indicated below)	High Distinction (HD) 85-100	Distinction (D) 75-84	Credit (C) 65-74	Pass (P) 50-64	Fail (F) 0-49
Criteria 1 Technical Skills (40 marks)	The tasks were appropriately completed, as is evident from the screenshots and analysis.	The tasks were appropriately completed, as is evident from the screenshots and analysis. Minor omissions only.	The tasks were mostly appropriately completed, as is evident from the screenshots and analysis. Some omissions.	The tasks were rightly done. Some evidence have been provided.	Little or no evidence of the tasks being appropriately done.
Criteria 2 Problem-Solving (15 marks)	Excellent demonstration of problem-solving skills as evidenced in the project log.	Very good demonstration of problem-solving skills as evidenced in the project log.	Good demonstration of problem-solving skills as evidenced in the project log.	Satisfactory demonstration of problem-solving skills as evidenced in the project log.	Unsatisfactory demonstration of problem-solving skills as evidenced in the project log/Missing project log.
Criteria 3 Report Writing (20 marks)	Excellent demonstration of report writing skills as evidenced in the report submission.	Very good demonstration of report writing skills as evidenced in the report submission.	Good demonstration of report writing skills as evidenced in the report submission.	Satisfactory demonstration of report writing skills as evidenced in the report submission.	Unsatisfactory demonstration of report writing skills as evidenced in the report submission.
Criteria 4 Subject Knowledge (15 marks)	Excellent demonstration of subject knowledge as evidenced in the short-answers and report content.	Very good demonstration of subject knowledge as evidenced in the short-answers and report content.	Good demonstration of subject knowledge as evidenced in the short-answers and report content.	Satisfactory demonstration of subject knowledge as evidenced in the short-answers and report content.	Unsatisfactory demonstration of r subject knowledge as evidenced in the short-answers and report content.
Criteria 5 Teamwork (5 marks)	Excellent demonstration of teamwork skills as evidenced in the team log.	Very good demonstration of teamwork skills as evidenced in the team log.	Good demonstration of teamwork skills as evidenced in the team log.	Satisfactory demonstration of teamwork skills as evidenced in the team log.	Unsatisfactory demonstration of teamwork skills/Missing details in the team log/No team log submitted.

Criteria 6 Spelling, Grammar, Presentation and Referencing (5 marks)	Excellent presentation, well written report with no spelling, grammatical and referencing mistakes	Very good report presentation, with minor spelling, minimal grammatical and referencing mistakes	Good presentation, with minor spelling, grammatical and referencing mistakes	Satisfactory presentation, with several spelling, grammatical and referencing mistakes	Major spelling, grammar, presentation and referencing mistakes
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